From Physical Properties to CUT Feasibility

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Nov 2nd, 2026

Property-Grounded CUT Affordance Reasoning

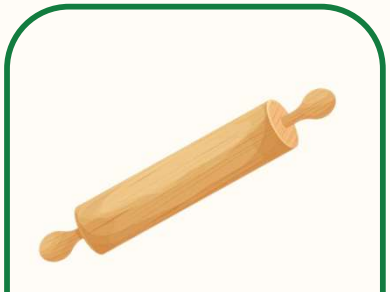
Ontological Goal

We evaluate whether a candidate can perform CUT on a target using explicitly represented physical properties, rather than object names or conventional functions.

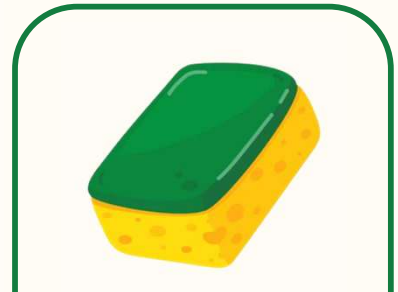




1. Plastic Card



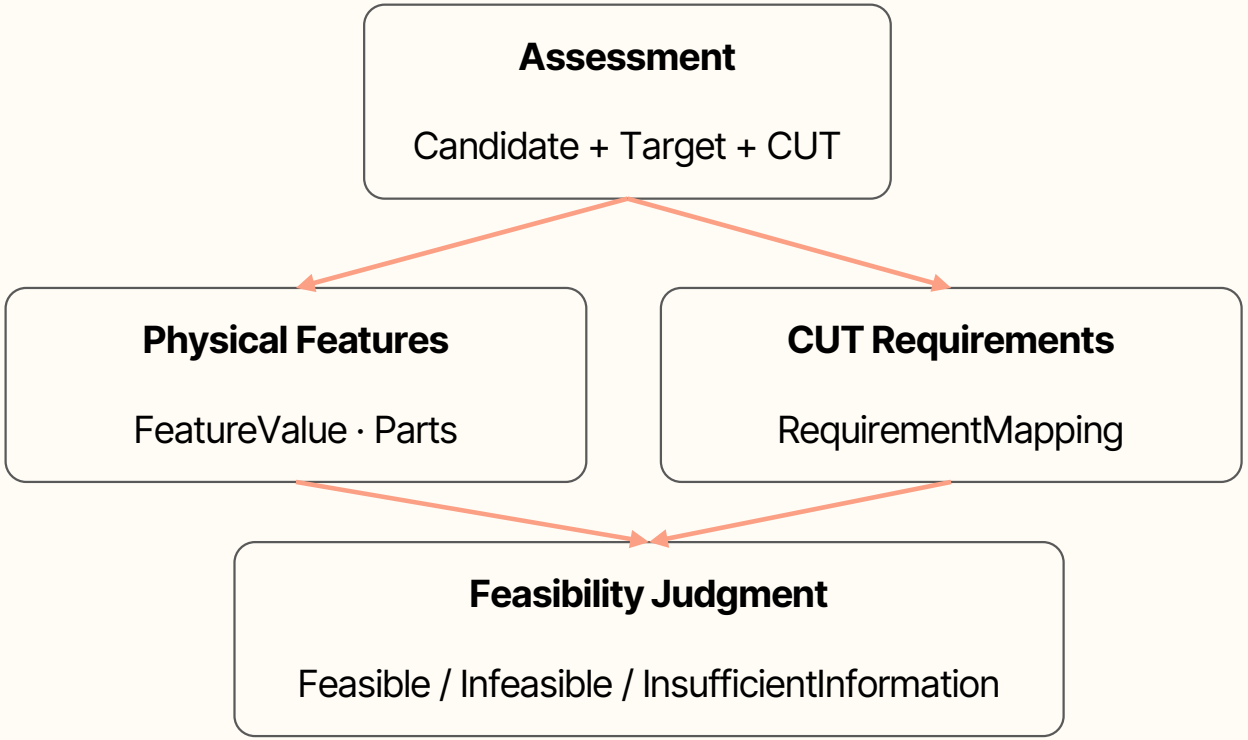
2. Rolling Pin



3. Sponge

To make this judgment, we first need to represent the relevant physical conditions!

How Our Ontology Represents CUT



- An **Assessment** links a candidate, target, and CUT action.
- **Physical features** describe the objects, while **RequirementMappings** specify what CUT requires under the target conditions.
- Together, they support requirement checking and feasibility judgments.

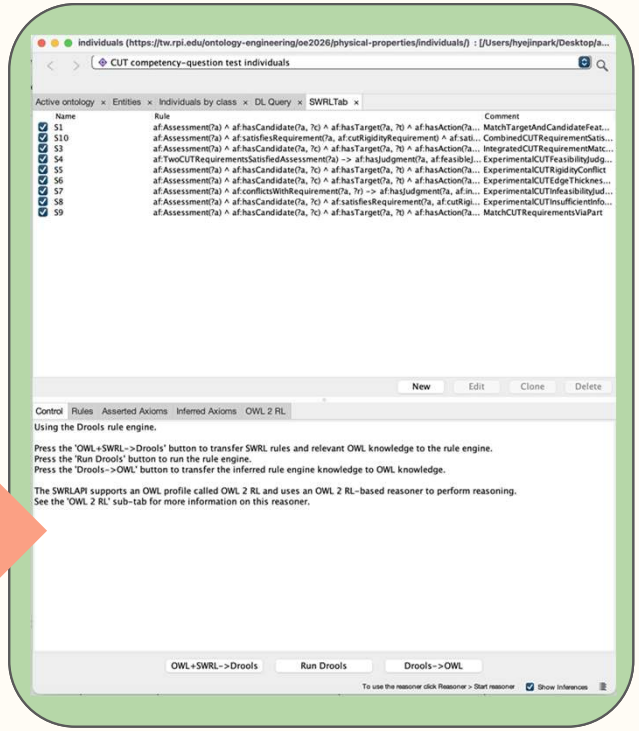
But how did this model evolve to answer our competency questions?

How Our Ontology Evolved

OE8 — Foundation

- Physical Features & RequirementMappings
- Assessment & Part relationships
- 2 initial SWRL rules

New Requirement:
Answer CQs with SPARQL



OE9 — Expanded Reasoning

- 9 SWRL rules
- Conflicts & final judgments
- Part-based inference
- Explicit missing information

- **OE8:** Established the knowledge structure and initial reasoning for requirement satisfaction.
- **OE9:** Extended SWRL from 2 to 9 rules to support queryable judgments, conflicts, and part-level reasoning.

With these extensions in place, we can now examine how the ontology supports answering our CQs through SPARQL.

From Reasoning to SPARQL Answers

Example: CUT on a high-hardness carrot For this target, CUT requires high rigidity and low edge thickness.

CQ1 — Can a plastic card cut a carrot, and why?
Direct comparison of feature levels

Rigidity	Satisfied
Edge Thickness	Satisfied

CQ2 — SWRL-Inferred Judgment
Which candidate can cut the carrot:
a card, rolling pin, or sponge?

Card	Feasible
Rolling Pin	Infeasible
Sponge	Infeasible

- **CQ1:** Using the existing feature values and requirement mappings, SPARQL directly confirms that the card satisfies both CUT requirements.
- **CQ2:** Using the expanded SWRL rules, SPARQL retrieves the inferred judgments: the card is feasible, while the rolling pin and sponge are infeasible.
- **Cross-validation:** A separate SPARQL query directly computes the same three CQ2 judgments from asserted data.
- **CQ3:** Knife vs. Scissors: Compare inferred CUT feasibility and OWL-based part structures.

Next, we'll demonstrate the actual queries and results for each competency question.

CQ1 - Can a rigid plastic card cut a carrot, and why?

SPARQL:

```
PREFIX af: <https://tw.rpi.edu/ontology-engineering/oe2026/physical-properties/>

SELECT ?candidate ?target ?action
    ?targetFeature ?targetLevel
    ?requirement ?requiredFeature ?requiredLevel
    ?candidateLevel ?requirementStatus
WHERE {
 VALUES ?assessment { af:cardCarrotCutAssessment }

 ?assessment af:hasCandidate ?candidate ;
        af:hasTarget ?target ;
        af:hasAction ?action .

?ac
?ta
?ta
?m
?requirement af:concernsFeature ?requiredFeature .
```

Snap SPARQL Query:

```
PREFIX af: <https://tw.rpi.edu/ontology-engineering/oe2026/physical-properties/>

SELECT ?judgment
WHERE {
  af:cardCarrotCutAssessment af:hasJudgment ?judgment .
}
```

Execute

?judgment
af:feasibleJudgment

```
OPTIONAL {
                                    ue ?candidateFeatureValue .
                                    asFeatureType ?requiredFeature ;
                                    ndidateLevel .
                                    ,
                                    iredLevel,
}
ORDER BY ?requirement
```

Result (partial)

candidate	requirement	requiredFeature	required Level	candidate Level	requirement Status
card	cutEdgeThicknessRequirement	edgeThickness	**low**	**low**	**Satisfied**
card	cutRigidityRequirement	rigidity	**high**	**high**	**Satisfied**

Answer: **Yes**. A rigid plastic card can cut a carrot. Card can cut a carrot because the required features for cutting a carrot have required levels that match the candidate card's feature levels.

REASONING: Asserted RDF triples for the comparisons; optional inferred hasJudgment for the feasibilityStatus column.
RUN IN: Standard SPARQL Query tab, SNAP SPARQL Qyery + HermiT Reasoner, with BOTH ontologies loaded.

CQ2 — Which available object can cut the carrot?

Competency question **Given the same carrot, which of a rigid plastic card, a rolling pin, and a kitchen sponge satisfy the CUT requirements?**
Scenario A service robot must slice a carrot but no knife is available. It must choose from the objects within reach using their physical features — not their names — and explain its choice.

① WHAT THE ONTOLOGY KNOWS (asserted individuals)

Individual	Rigidity	Edge thickness
cardCandidate	high	low
rollingPinCandidate	high	high
spongeCandidate	low	high
carrotTarget	**hardness = high**	(target)

One Assessment per object — cardCarrotCutAssessment, rollingPinCarrotCutAssessment, spongeCarrotCutAssessment — each with the same target (carrotTarget) and action (cut).

② WHAT CUT REQUIRES FOR A HARD TARGET (main ontology)

Requirement mapping	If target has	Candidate needs
cutRigidityMapping	hardness = high	**rigidity = high**
cutEdgeThicknessMapping	hardness = high	**edge thickness = low**

③ SHARED EVALUATION RULE (use case §IV)

Per requirement: same level = satisfied · different level = conflict · value marked unspecified = unresolved

FEASIBLE
every requirement satisfied

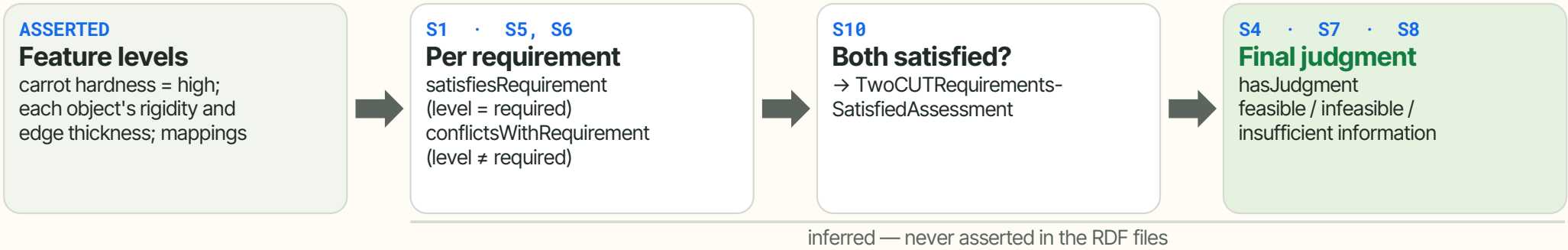
INFEASIBLE
at least one conflict

INSUFFICIENT INFO
unresolved, no conflict

Expected answer **Only the card.** The rolling pin's edge is too thick; the sponge is neither rigid enough nor thin-edged. Missing information must never be treated as a conflict.

CQ2 — The answer comes from entailment

INFERENCE CHAIN (SWRL rules, run by HermiT — OWL 2 DL + SWRL)



Q2.1 · MAIN CQ2 QUERY

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX af: <https://tw.rpi.edu/ontology-engineering/oe2026/physical-properties/>
SELECT ?candidate ?judgment
WHERE {
  ?assessment rdf:type af:Assessment .
  ?assessment af:hasTarget af:carrotTarget .
  ?assessment af:hasAction af:cut .
  ?assessment af:hasCandidate ?candidate .
  ?assessment af:hasJudgment ?judgment .    # inferred
  FILTER (?candidate = af:cardCandidate ||
    ?candidate = af:rollingPinCandidate || ?candidate = af:spongeCandidate)
} ORDER BY ?candidate
```

SAME QUERY, TWO WAYS TO RUN IT

Standard SPARQL Query tab
asserted triples only
→ 0 rows

Snap SPARQL plugin + HermiT
OWL 2 DL entailment incl. SWRL rules
→ 3 rows: the three judgments

Open Individuals .rdf → Reasoner ▸ HermiT ▸ Start → Snap SPARQL tab

CQ2 — Results: only the card can cut the carrot

ANSWER **Card → Feasible. Rolling pin → Infeasible. Sponge → Infeasible.** Matches the expected answer.

Q2.1 + Q2.2 · INFERRED JUDGMENT AND EXPLANATION (Snap SPARQL + HermiT)

Candidate	satisfies (S1)	conflicts with (S5, S6)	hasJudgment
card	rigidity, edge thickness	—	**feasible**
rolling pin	rigidity	edge thickness (high)	**infeasible**
sponge	—	rigidity (low), edge thickness (high)	**infeasible**

The sponge is a conflict, not missing information: its deformability is represented as low rigidity. Without a reasoner, Q2.1 and Q2.2 return 0 rows.

LIMITATIONS → NEXT STEPS

- Conflict rules S5/S6 test specific levels — high/medium/low are not declared distinct (fix: owl:AllDifferent + one generic rule).
- Insufficient information needs an explicit hasUnspecifiedFeatureType marker (open-world semantics).
- Requirement mappings cover only high-hardness targets.

VALIDATION

Q2.3 · no reasoner (standard tab)
Recomputes the verdicts from asserted feature levels:
FEASIBLE / INFEASIBLE / INFEASIBLE ✓

Q2.4 · all carrot assessments → 8 rows

FEASIBLE card · knife · scissors · testCandidateA

INFEASIBLE rolling pin · sponge · testCandidateB

INSUFFICIENT INFO testCandidateC (edge thickness unspecified)

Test A/B have the card's / rolling pin's levels under neutral names → same judgments: features decide, not names.

CQ3 Same CUT outcome, different part structures

Can a knife and scissors both meet CUT requirements for the same carrot, and how do their part structures differ?

RESULT Both tools are feasible; scissors have multipart and interacting structure.

EVIDENCE | Verified Q3.6 + Q3.4 results

Candidate	CUT judgment	Structural classes
Knife	Feasible	Neither queried class inferred
Scissors	Feasible	MultiPartObject InteractingPartObject

Controls: testCandidate3 → multipart;
testCandidate4 → both classes.

REASONING

High rigidity + low edge thickness → S9 → S10 / S4 → feasible
Distinct parts + interaction → OWL structural classifications

SPARQL | Q3.6 core query

```
SELECT ?candidate ?requirement ?judgment
WHERE {
 ?a a af:Assessment ;
    af:hasCandidate ?candidate ;
    af:hasTarget af:carrotTarget ;
    af:hasAction af:cut ;
    af:satisfiesRequirement ?requirement ;
    af:hasJudgment ?judgment .
 FILTER (?candidate = af:knifeCandidate ||
         ?candidate = af:scissorsCandidate)
}
```

4 rows: two satisfied requirements per tool.
Project prefix, full queries and results in notes.

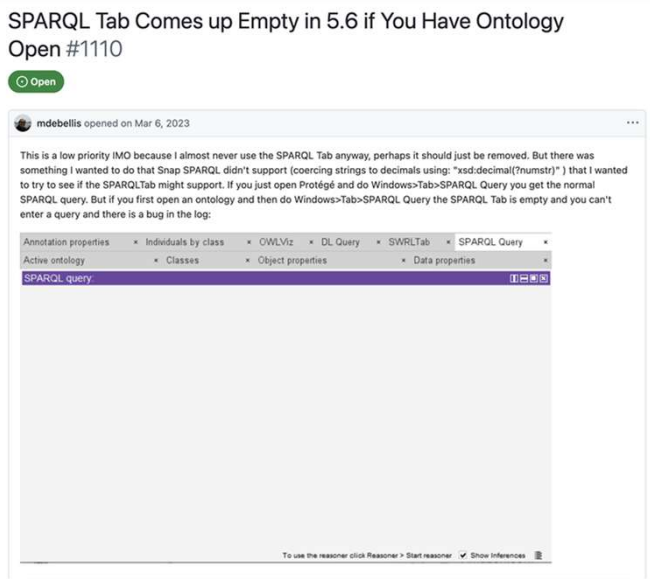
Knife: non-inference does not prove nonmembership. | Snap SPARQL + HermiT; Show Inferences on.

How AI Supported Our Ontology Development

1. Ontology Modeling & Reasoning — ChatGPT
   OWL restrictions, SWRL rules, class relationships, and reasoning behavior review.
   Examples: requirement satisfaction, open-world semantics, inconsistent judgments, RDF hygiene.

2. SPARQL Development & Validation — ChatGPT / Claude
   Assisted with query construction, debugging, refinement, and interpretation of results.
   Examples: CQ2 query refinement with Claude; CQ3 query troubleshooting and validation discussions with ChatGPT.

3. Technical Troubleshooting
   Investigate a Protégé SPARQL plugin compatibility issue,
   evaluate a patched plugin workaround,
   and consider Snap SPARQL.
   GitHub issue #1110 · Protégé SPARQL tab · Snap SPARQL



Website Walkthrough

Webpage: About | Action Feasibility from Physical Properties

Current Version:

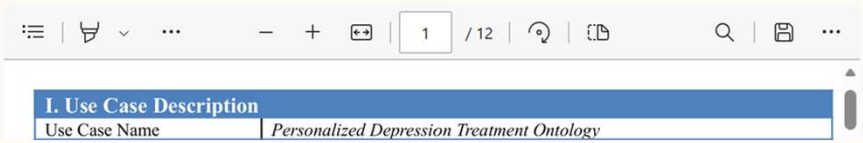
- Downlink for documents has attached;



- All files been uploaded;
- Other parts like Related Works and Getting Involved will be filled in future weeks.

Goal for the Final Version:

- Match with previous years, preview for documents will be available;



- Reorganized the structure of some pages to improve readability;
- ……